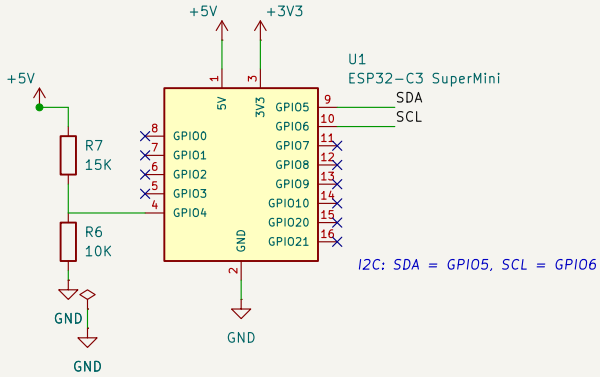
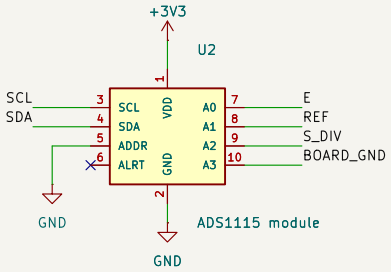
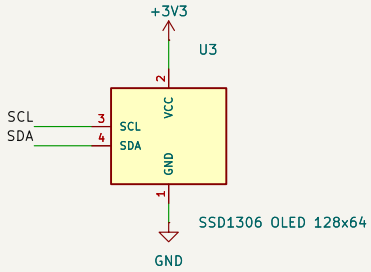
pH electrode simulator and leakage tester

1. Power the ADS1115 and the OLED from 3.3 V, never from 5 V.
2. The simulator voltage E is measured BEFORE R2; the board input VIN is worked out from S.
3. Leakage: I = (VIN − E) / R2 after a two-point calibration with SW2 pressed.
4. The simulator floats and REF has no divider of its own: channel A is valid only with a pH board connected.
5. Optional, not fitted: 10k from 3.3 V and 10k to GND with 100 nF, midpoint on REF, make A valid on its own.



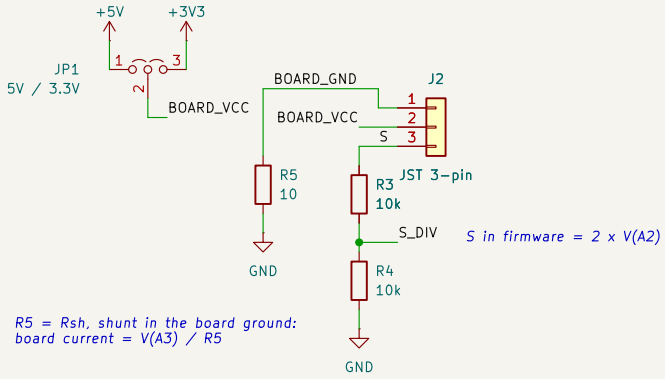


ADDR to GND: address 0x48
A0 − A1: simulator voltage E
A2: S / 2
A3: voltage on the shunt R5



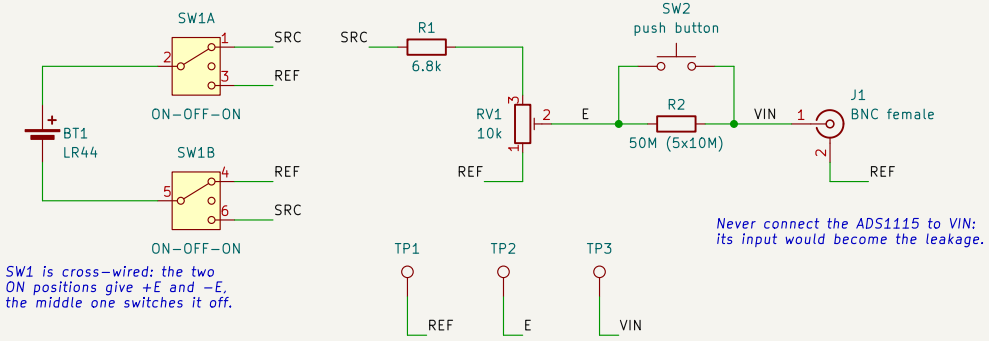
Address 0x3C. Wire by the printed pin names.

J2 as built: 1 GND (through R5), 2 supply (JP1: 5 V or 3.3 V), 3 S.
P2 on the pH board is 1 VCC, 2 com, 3 S: the cable is crossed.
Wire it by function and check it before power-up.



R5 = Rsh, shunt in the board ground:
board current = V(A3) / R5

SW2 pressed: R2 shorted, the board input is E (calibration).
SW2 released: the signal goes through R2 (leakage measurement).



SW1 is cross-wired: the two
ON positions give +E and −E,
the middle one switches it off.

Never connect the ADS1115 to VIN:
its input would become the leakage.

Test pins: TP1 BNC shield (REF), TP2 BNC + before R2 (E),
TP3 BNC + after R2 (VIN). TP3 is the high-impedance node:
touch it only with an electrometer.

Hardware licence: CERN-OHL-S-2.0
Wiring as in docs/wiring.png; firmware src/main.cpp.
igorlab

Sheet: /
File: calibrator.kicad_sch

Title: pH electrode simulator and leakage tester

Size: A4	Date: 2026-10-21	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/1